

Junior AI Investigator

A case-review workbench for long-term-care referrals. It orders the queue, shows the evidence, and keeps the investigator in control.

THE PRODUCT

- User: a fraud investigator starting a morning queue.
- Job: decide which referrals need a full investigation, one focused check, or possible false-positive closure.
- Built: a 50-case queue with three triage lanes, case briefs, evidence, next steps, questions, and a decision log.
- Not built: authentication, a database, multi-user workflow, or claims-system integration. The CSV is the input.
- No case receives a fraud probability because the file has no outcome labels.

WHAT THE DATA SHOWS

- Across the queue, pairwise Spearman correlations range from 0.62 to 0.89.
- The first principal component explains 74.5% of variance; the equal-weight index correlates 0.999 with it.
- Equal weights are used because there are no outcomes for supervised weighting. Cases can still move when weights change, so sensitivity is reported.
- Two large gaps form groups of 30, 12, and 8. Safety rules produce final lanes of 29, 13, and 8.

Spearman 0.62-0.89 | PC1 74.5% | index-PC1 0.999 | final lanes 29 / 13 / 8

 Priority

 Judgment

 Likely false positive

System design and human control

SYSTEM DESIGN

- triage.py sets the lane, evidence, numbers, and first check. These steps are deterministic.
- Before the shift, the model drafts the headline, summary, ordinary explanation, and missing records for all 50 cases: 44 passed first time and 6 after repair.
- A frozen 19-case regression comparison found that model-selected evidence missed a required signal in 11 cases. Engine-selected evidence covers all triggered flags and care-setting clashes.
- Drafts are checked against the case facts. A failed draft gets one repair attempt, then the rule brief is used.
- Cached briefs are tied to the fact sheet and checked again when loaded. No API key is needed for the demo.
- At scale, reference values should come from a fixed calibration window by care type and run inside an approved environment.

HUMAN CONTROL

- The investigator records agree, disagree, or not sure for each finding.
- A recommendation never takes action. An override requires a written reason.
- Questions about paying or denying a claim are refused before they reach the model.
- Profile-group retention is shown as sensitivity: 46 of 50 cases retain their group in at least 90% of every stress test. It is not a correctness probability.
- The audit gate reads 2 targeted and 14 random cases. If review would detect a sampled problem, 14 clean random reads give a 90% upper bound of 2 hidden problems in the 27-case lot. Any adverse decision blocks bulk closing.
- Every decision logs the model, prompt version, displayed text, and fact fingerprint.

Limits, risks and next tests

LIMITS AND CONTROLS

- No outcomes means no measured accuracy and no fraud probability.
- All percentiles, groups, and ranges describe this referred queue of 50 cases.
- Generated text can sound too certain. Deterministic evidence, text checks, and the rule fallback limit that risk.
- The 10% audit tolerance and 90% confidence level are prototype policy choices. Operations and compliance should set them in production.
- Claim data requires an approved model endpoint. Human review and the decision log reduce automation bias.
- The prototype leaves out authentication, a production datastore, and claims-system integration.

NEXT TESTS AND ASSUMPTIONS

- The assignment mentions nine signals; the CSV contains ten. All ten are used.
- Zero means normal only for the four flags. Continuous measures use a descriptive benign-profile range and are treated as one-sided.
- Claim amount is exposure. It orders work within a lane but does not enter the index.
- Run beside current practice and measure time to decision plus audited misses.
- Compare model and rule briefs in a blinded investigator review of clarity and actionability.
- For production, freeze and version calibration values by care type, then monitor drift.